EMF & FARADAY'S LAW

Transformer EMF *stationary loop, B changes*

$$V_{\text{emf}} = -\int_S \frac{\partial \mathbf{B}}{\partial t} \cdot d\mathbf{s}$$

Motional EMF *loop moves, B static*

$$V_{\text{emf}} = \oint (\mathbf{v} \times \mathbf{B}) \cdot d\mathbf{l}$$

General (flux form) *always works*

$$V_{\text{emf}} = -\frac{d\Phi}{dt} \qquad \Phi = \int_S \mathbf{B} \cdot d\mathbf{s}$$

N-turn coil *transformer, N turns*

$$V_{\text{emf}} = -N\frac{d\Phi}{dt} = -\frac{d\lambda}{dt} \quad \lambda = N\Phi$$

Uniform B shortcut *B same everywhere on loop*

$$V_{\text{emf}} = -\frac{\partial B}{\partial t} \cdot A$$

Long wire + rect loop *wire near loop, r: a → b, height c*

$$V_{\text{emf}} = -\int_0^c \int_a^b \frac{\partial}{\partial t}\left(\frac{\mu_0 I}{2\pi r}\right) dr\, dz$$

SURFACE ELEMENT ds

$d\mathbf{s} = \hat{n}\, da$ *RHR: curl fingers in loop dir., thumb* $= \hat{n}$

Loop plane	$d\mathbf{s}$
x-y	$\hat{z}\, dx\, dy$ or $\hat{z}\, r\, dr\, d\phi$
y-z	$\hat{x}\, dy\, dz$
x-z	$\hat{y}\, dx\, dz$
coplanar/toroid	$\hat{\phi}\, dr\, dz$

Uniform B: $\iint da = A$ *(just the loop area)*

DOT PRODUCTS $\hat{a} \cdot \hat{b}$

Same dir.= 1; perpendicular= 0

	$\hat{x}$	$\hat{y}$	$\hat{z}$	$\hat{r}$	$\hat{\phi}$
$\hat{x}$	1	0	0	0	0
$\hat{y}$	0	1	0	0	0
$\hat{z}$	0	0	1	0	0
$\hat{\phi}$	0	0	0	0	1

CROSS PRODUCT $\mathbf{A} \times \mathbf{B}$

$$\mathbf{A} \times \mathbf{B} = \begin{vmatrix} \hat{x} & \hat{y} & \hat{z} \\ A_x & A_y & A_z \\ B_x & B_y & B_z \end{vmatrix}$$

$= (A_y B_z - A_z B_y)\hat{x} - (A_x B_z - A_z B_x)\hat{y} + (A_x B_y - A_y B_x)\hat{z}$

Plane wave ($\partial_x = \partial_y = 0$): only ∂_z survives in curl.

B FIELDS

Infinite wire *single wire, N = 1*

$$\mathbf{B} = \hat{\phi}\, \frac{\mu_0 I}{2\pi r}$$

Inside toroid *N-turn coil*

$$\mathbf{B} = \hat{\phi}\, \frac{\mu N I}{2\pi r} \qquad \mu = \mu_r \mu_0$$

*N in $\mathbf{B}$ only if the **source** is multi-turn. In $V_{emf} = -N d\Phi/dt$, N is the **receiver** turns.*

$\partial B/\partial t$ TABLE

$B(t)$	$\partial B/\partial t$
$B_0 \sin(\omega t)$	$B_0 \omega \cos(\omega t)$
$B_0 \cos(\omega t)$	$-B_0 \omega \sin(\omega t)$
$B_0 e^{-\alpha t}$	$-\alpha B_0 e^{-\alpha t}$
B_0 (const)	0

SIGN → CURRENT DIRECTION

V_{emf}	Inside source	I dir.
< 0	$-$ → $+$ terminal	$-\hat{\phi}$ (CW)
> 0	$+$ → $-$ terminal	$+\hat{\phi}$ (CCW)
$= 0$	—	none

Lenz: induced I opposes the change in Φ.

ln INTEGRAL

$B \propto 1/r$, loop spans $r = a$ to $r = b$

$$\int_a^b \frac{dr}{r} = \ln\frac{b}{a} = \ln b - \ln a$$

$\ln 2 = 0.693 \quad \ln 3 = 1.099 \quad \ln 1.2 = 0.182$

ROTATING LOOP — GENERATOR

$$\Phi = AB_0 \cos(\omega t + \phi_0)$$

$$V_{\text{emf}} = AB_0 \omega \sin(\omega t + \phi_0)$$

$$I_{\text{amp}} = \frac{AB_0\omega}{R} \qquad \omega = 2\pi f = 2\pi \frac{\text{rpm}}{60}$$

SI PREFIXES

Prefix	Sym	Val	Prefix	Sym	Val
pico	p	10^{-12}	kilo	k	10^3
nano	n	10^{-9}	mega	M	10^6
micro	μ	10^{-6}	giga	G	10^9
milli	m	10^{-3}	tera	T	10^{12}

j IDENTITIES

Expr	Result	Expr	Result
j^2	-1	$j \times (-j)$	1
$1/j$	$-j$	$-1/j$	j
j^3	$-j$	j^4	1
$e^{-j\pi/2}$	$-j$	$e^{j\pi/2}$	j
$e^{j\pi}$	-1	$e^{j2\pi}$	1

DISPLACEMENT & CONDUCTION CURRENT

Conduction *resistive, $\sigma \neq 0$*

$$J_c = \sigma E \qquad I_c = J_c \cdot A \qquad R = \frac{d}{\sigma A}$$

Displacement *capacitive, changing E*

$$J_d = \varepsilon \frac{\partial E}{\partial t} \qquad I_d = J_d \cdot A \qquad C = \frac{\varepsilon_r \varepsilon_0 A}{d}$$

Lossy cap circuit *both σ and ε_r*

$$I = I_c + I_d = \frac{V}{R} + C\frac{dV}{dt}$$

CHARGE RELAXATION

$$\rho_v(t) = \rho_{v0}\, e^{-t/\tau_r} \qquad \tau_r = \frac{\varepsilon_r \varepsilon_0}{\sigma}$$

Larger τ_r = slower dissipation = better insulator.

PHASOR METHOD (E → H)

1. $\mathbf{E}(z,t) = \text{Re}\{\tilde{\mathbf{E}}(z)\, e^{j\omega t}\}$
2. Strip $e^{j\omega t}$, keep e^{-jkz}:
$\cos(\omega t - kz) \to e^{-jkz} \quad \sin(\omega t - kz) \to -j\, e^{-jkz}$
3. Faraday in phasor (plane wave: only $\partial/\partial z$ survives):

$$\tilde{\mathbf{H}} = -\frac{1}{j\omega\mu} \nabla \times \tilde{\mathbf{E}}$$

4. Back to time: $\text{Re}\{Ae^{j\phi}e^{j(\omega t - kz)}\} = A\cos(\omega t - kz + \phi)$

UNIT CIRCLE

Angle	Deg	cos	sin	Q
0	0	1	0	—
$\pi/6$	30	$\sqrt{3}/2$	$1/2$	1
$\pi/4$	45	$\sqrt{2}/2$	$\sqrt{2}/2$	1
$\pi/3$	60	$1/2$	$\sqrt{3}/2$	1
$\pi/2$	90	0	1	—
$2\pi/3$	120	$-1/2$	$\sqrt{3}/2$	2
$3\pi/4$	135	$-\sqrt{2}/2$	$\sqrt{2}/2$	2
$5\pi/6$	150	$-\sqrt{3}/2$	$1/2$	2
π	180	-1	0	—
$7\pi/6$	210	$-\sqrt{3}/2$	$-1/2$	3
$5\pi/4$	225	$-\sqrt{2}/2$	$-\sqrt{2}/2$	3
$4\pi/3$	240	$-1/2$	$-\sqrt{3}/2$	3
$3\pi/2$	270	0	-1	—
$5\pi/3$	300	$1/2$	$-\sqrt{3}/2$	4
$7\pi/4$	315	$\sqrt{2}/2$	$-\sqrt{2}/2$	4
$11\pi/6$	330	$\sqrt{3}/2$	$-1/2$	4
2π	360	1	0	—

FIELDS

$\mathbf{E}$ — electric field (V/m)
$\mathbf{B}$ — mag. flux density (T)
$\mathbf{H}$ — mag. field (A/m)
ρ_v — charge density (C/m^3)
$\Phi = \int_S \mathbf{B} \cdot d\mathbf{s}$ — flux (Wb)
$\lambda = N\Phi$ — flux linkage (Wb)

CURRENTS

$J_c = \sigma E$ — cond. density (A/m^2)
$J_d = \varepsilon\, \partial E/\partial t$ — disp. density
$I_c = J_c \cdot A$ — cond. current (A)
$I_d = J_d \cdot A$ — disp. current (A)
$R = d/(\sigma A)$ — resistance (Ω)
$C = \varepsilon_r \varepsilon_0 A/d$ — capacitance (F)
$\tau_r = \varepsilon_r \varepsilon_0/\sigma$ — relax. time (s)

MATERIALS

$\mu_0 = 4\pi \times 10^{-7}$ H/m
μ_r — rel. permeability
$\mu = \mu_r \mu_0$ — permeability
$\varepsilon_0 \approx \frac{1}{36\pi} \times 10^{-9}$ F/m
ε_r — rel. permittivity
$\varepsilon = \varepsilon_r \varepsilon_0$ — permittivity
σ — conductivity (S/m)

CIRCUIT & WAVE

V — voltage (V)
R — resistance (Ω)
C — capacitance (F)
$\omega = 2\pi f$ — ang. freq (rad/s)
$k = \omega\sqrt{\mu\varepsilon}$ — wavenumber
$\lambda = 2\pi/k$ — wavelength (m)
$c = 3 \times 10^8$ m/s

PHASOR & GEOMETRY

$\eta = \sqrt{\mu/\varepsilon}$ — imp. (Ω)
$\eta_0 = 120\pi \approx 377\,\Omega$
$k/(\omega\mu) = 1/\eta$
$j^2 = -1$; $\tilde{\mathbf{E}}$: drop $e^{j\omega t}$
A — area (m^2); d — sep. (m)
N — turns; r — radial (m)

PLANE WAVE PARAMETERS (LOSSLESS)

Angular frequency *ω: rad/s; f: Hz*

$$\omega = 2\pi f \qquad f = \frac{\omega}{2\pi}$$

Phase velocity *m/s; nonmagnetic: $\mu_r = 1$*

$$u_p = \frac{c}{\sqrt{\varepsilon_r}} \qquad c = 3 \times 10^8 \text{ m/s}$$

Wavenumber k *rad/m*

$$k = \frac{\omega}{u_p} = \frac{2\pi}{\lambda} = \frac{\omega\sqrt{\varepsilon_r}}{c}$$

Wavelength & impedance *m & Ω*

$$\lambda = \frac{u_p}{f} = \frac{2\pi}{k} \text{ (m)} \qquad \eta = \frac{\eta_0}{\sqrt{\varepsilon_r}} \ (\Omega)$$

Order: $\omega \to u_p \to k \to \lambda \to \eta$.

LOSSLESS PLANE WAVE — E AND H

General E

$$\mathbf{E} = E_0 \cos(\omega t - k\,\hat{k}\cdot\mathbf{r} + \phi_0)\,\hat{e}$$

$\hat{k}$: travel dir.; $\hat{e}$: polarization dir.; $\hat{k}\cdot\mathbf{r}$ picks the axis.
H from E *$\hat{k}$, $\hat{E}$, $\hat{H}$ form right-hand set*

$$\mathbf{H} = \frac{1}{\eta}\,(\hat{k} \times \mathbf{E})$$

Phasor form ($+z$ **propagation**)

$$\tilde{\mathbf{E}} = E_0 e^{j\phi_0} e^{-jkz}\hat{e} \qquad \tilde{\mathbf{H}} = \frac{1}{\eta}\hat{k} \times \tilde{\mathbf{E}}$$

DIRECTION OF PROPAGATION

Arg. contains	$\hat{k}$	Arg. contains	$\hat{k}$
$\omega t - kz$	$+\hat{z}$	$\omega t + kz$	$-\hat{z}$
$\omega t - ky$	$+\hat{y}$	$\omega t + ky$	$-\hat{y}$
$\omega t - kx$	$+\hat{x}$	$\omega t + kx$	$-\hat{x}$

Minus before k-axis $\Rightarrow$ + direction. Plus $\Rightarrow$ − direction.

$\hat{k} \times \hat{E}$ — FIND $\hat{H}$ DIRECTION

$\hat{k}$	$\hat{E}$	$\hat{H}$	$\hat{k}$	$\hat{E}$	$\hat{H}$
$+\hat{z}$	$\hat{x}$	$+\hat{y}$	$+\hat{z}$	$\hat{y}$	$-\hat{x}$
$+\hat{y}$	$\hat{x}$	$-\hat{z}$	$+\hat{y}$	$\hat{z}$	$+\hat{x}$
$+\hat{x}$	$\hat{y}$	$+\hat{z}$	$+\hat{x}$	$\hat{z}$	$-\hat{y}$
$-\hat{x}$	$\hat{y}$	$-\hat{z}$	$-\hat{x}$	$\hat{z}$	$+\hat{y}$
$-\hat{y}$	$\hat{x}$	$+\hat{z}$	$-\hat{z}$	$\hat{x}$	$-\hat{y}$

Cyclic: $\hat{x}\times\hat{y}=\hat{z}$, $\hat{y}\times\hat{z}=\hat{x}$, $\hat{z}\times\hat{x}=\hat{y}$. Reverse $\Rightarrow$ flip sign.

TRIG PHASE SHIFTS *($\theta \equiv \omega t - kz$)*

Input	Result	Input	Result
$\sin(\theta+\pi/2)$	$\cos\theta$	$\cos(\theta+\pi/2)$	$-\sin\theta$
$\sin(\theta-\pi/2)$	$-\cos\theta$	$\cos(\theta-\pi/2)$	$\sin\theta$
$\sin(\theta\pm\pi)$	$-\sin\theta$	$\cos(\theta\pm\pi)$	$-\cos\theta$

POLARIZATION — GENERAL WAVE

$$\mathbf{E} = a_x \cos(\omega t - kz)\hat{x} + a_y \cos(\omega t - kz + \delta)\hat{y}$$

a_x, a_y: bounding box half-widths (V/m); δ: phase diff (rad)
*$\delta = 0$ or $\pi \Rightarrow$ **linear**; $a_x = a_y$, $\delta = \pm\pi/2 \Rightarrow$ **circular***
Auxiliary angle $\psi_0 = \arctan(a_y/a_x) \in (0, \pi/2)$ **Rotation angle** γ **(major axis from $\hat{x}$)** $\gamma \in [0, \pi)$

$$\tan(2\gamma) = \tan(2\psi_0)\cos\delta$$

If sign(γ) $\neq$sign(δ): $\gamma \leftarrow \gamma \pm \frac{\pi}{2}$. **Ellipticity angle** χ $\chi \in [-\pi/4, \pi/4]$

$$\sin(2\chi) = \sin(2\psi_0)\sin\delta$$

Axial ratio $R = 1/\tan|\chi|$ *$R \geq 1$; $R = 1$: circular; $R \to \infty$: linear*

χ SHAPE & HANDEDNESS

χ (rad)	Polarization Type	Handedness
0	Linear	None
$(0,\ \pi/4)$	RHEP	RH (CCW)
$\pi/4$	RHC	RH (CCW)
$-\pi/4$	LHC	LH (CW)
$(-\pi/4,\ 0)$	LHEP	LH (CW)

QUADRANT RULE FOR $\tan(2\gamma)$

$N = \sin(2\psi_0)\cos\delta$, $\quad D = \cos(2\psi_0)$:

N	D	2γ (deg)	2γ (rad)
+	+	$\arctan(N/D)$	$\arctan(N/D)$
+	−	$180 - \arctan\lvert N/D\rvert$	$\pi - \arctan\lvert N/D\rvert$
−	−	$180 + \arctan\lvert N/D\rvert$	$\pi + \arctan\lvert N/D\rvert$
−	+	$360 - \arctan\lvert N/D\rvert$	$2\pi - \arctan\lvert N/D\rvert$

$\gamma = (2\gamma)/2$. $\quad D = 0 \Rightarrow 2\gamma = \frac{\pi}{2} \Rightarrow \gamma = \frac{\pi}{4}$.

PHASOR $\leftrightarrow$ TIME DOMAIN

Phasor coeff	Time domain ($\times e^{-\alpha z}$ in lossy)
A (real)	$A\cos(\omega t - \beta z)$
$-jA$	$A\sin(\omega t - \beta z)$
$+jA$	$-A\sin(\omega t - \beta z)$
$Ae^{j\phi}$	$A\cos(\omega t - \beta z + \phi)$
$A + jB$	$\lvert Z\rvert\cos(\omega t - \beta z + \theta)$

$|Z| = \sqrt{A^2 + B^2}$, $\theta = \arctan(B/A)$ (check quadrant)

AVERAGE POWER DENSITY (POYNTING)

General *W/m²*

$$\mathbf{S}_{\text{av}} = \tfrac{1}{2}\,\text{Re}\{\tilde{\mathbf{E}} \times \tilde{\mathbf{H}}^*\}$$

Lossless shortcut $|E_0|^2 = a_x^2 + a_y^2 + a_z^2$

$$\mathbf{S}_{\text{av}} = \frac{|E_0|^2}{2\eta}\,\hat{k}$$

Direction= $\hat{k}$. Lossy: $\mathbf{S}_{\text{av}} = \frac{|E(0)|^2}{2|\eta_c|}e^{-2\alpha z}\cos\theta_\eta\,\hat{k}$.

LOSSY MEDIA — CLASSIFICATION

Loss tangent

$$\frac{\varepsilon''}{\varepsilon'} = \frac{\sigma}{\omega\varepsilon_r\varepsilon_0}$$

$\varepsilon''/\varepsilon'$	Type	Use
> 100	Good conductor	GC approx
0.01 to 100	Quasi-conductor	Exact eqs
< 0.01	Low-loss diel.	LD approx

$\varepsilon' = \varepsilon_r\varepsilon_0$; $\varepsilon'' = \sigma/\omega$; nonmag.: $\mu = \mu_0$.

ANY MEDIUM — EXACT ($X = \sqrt{1 + (\varepsilon''/\varepsilon')^2}$)

$$\alpha = \omega\sqrt{\frac{\mu\varepsilon'}{2}}\sqrt{X-1}\ \text{Np/m}, \quad \beta = \omega\sqrt{\frac{\mu\varepsilon'}{2}}\sqrt{X+1}\ \text{rad/m}$$

$$\eta_c = \frac{\eta}{X^{1/2}}e^{j\theta_\eta}, \quad \theta_\eta = \tfrac{1}{2}\arctan\tfrac{\sigma}{\omega\varepsilon'}\ \text{(rad)}$$

$\eta = \eta_0/\sqrt{\varepsilon_r}$; $\lambda = 2\pi/\beta$; $u_p = \omega/\beta$.

LOSSLESS ($\sigma = 0$)

$\alpha = 0 \quad \beta = k = \omega\sqrt{\mu\varepsilon}$ (rad/m)
$\eta_c = \eta = \sqrt{\mu/\varepsilon}$ (Ω, real; $\theta_\eta = 0$)
$u_p = c/\sqrt{\varepsilon_r}$ (m/s); **E** and **H** in phase

LOW-LOSS MEDIUM ($\varepsilon''/\varepsilon' \ll 1$)

$$\alpha \approx \tfrac{\sigma}{2}\sqrt{\tfrac{\mu}{\varepsilon'}} = \tfrac{\sigma\eta}{2}\ \text{(Np/m)}$$

$$\beta \approx \omega\sqrt{\mu\varepsilon'} = \tfrac{\omega\sqrt{\varepsilon_r}}{c}\ \text{(rad/m)}$$

$$\eta_c \approx \eta = \sqrt{\tfrac{\mu}{\varepsilon'}}\ (\Omega,\ \angle \approx 0)$$

GOOD CONDUCTOR ($\varepsilon''/\varepsilon' \gg 1$)

$$\alpha \approx \beta \approx \sqrt{\pi f\mu\sigma}\ \text{(Np/m, rad/m)}$$

$$\eta_c \approx (1+j)\sqrt{\tfrac{\pi f\mu}{\sigma}} \approx (1+j)\tfrac{\alpha}{\sigma}\ (\Omega)$$

$$\delta_s = \frac{1}{\alpha} = \frac{1}{\sqrt{\pi f\mu\sigma}}\ \text{(m, good cond.)}$$

$\tilde{\mathbf{H}}$ PHASOR $\to$ $\tilde{\mathbf{E}}$ IN LOSSY MEDIA

Step 1 *read α, β from* $\tilde{\mathbf{H}} \propto e^{-\alpha y}e^{-j\beta y}$ **Step 2** $\eta_c = \frac{j\omega\mu_0}{\gamma} = \frac{\omega\mu_0(\beta + j\alpha)}{\alpha^2+\beta^2}$ **Step 3** $\tilde{\mathbf{E}} = -\eta_c(\hat{k} \times \tilde{\mathbf{H}})$ **Step 4** $Ae^{j\phi}e^{-\alpha y} \xrightarrow{\text{Re}\{\cdot e^{j\omega t}\}} Ae^{-\alpha y}\cos(\omega t - \beta y + \phi)$ *Convert each $A + jB \to |Z|\angle\theta$ first.*

USEFUL NUMBERS

$\sqrt{2} \approx 1.414$; $\sqrt{3} \approx 1.732$; $\sqrt{6} \approx 2.449$
$\sqrt{2.5} \approx 1.581$; $\sqrt{2.56} = 1.6$; $\sqrt{9} = 3$
$\arctan(4/3) \approx 53.1$; $\arctan(3/4) \approx 36.9$
$\arccos(2/3) \approx 48.2$; $\arctan(24/7) \approx 73.7$

WAVE PARAMS

$\omega = 2\pi f$ — ang. freq (rad/s)
$k = \omega\sqrt{\varepsilon_r}/c$ — wavenumber
$\lambda = 2\pi/k$ — wavelength (m)
$u_p = c/\sqrt{\varepsilon_r}$ — phase vel. (m/s)
$\eta = \eta_0/\sqrt{\varepsilon_r}$ — impedance (Ω)
$c = 3 \times 10^8$ m/s; $\eta_0 = 377\,\Omega$

LOSSLESS MEDIA

$\sigma = 0$ (ideal dielectric)
$\alpha = 0$; no attenuation
$\eta = \eta_0/\sqrt{\varepsilon_r}$ (real, Ω)
E and **H** in phase ($\theta_\eta = 0$)
$S_{\text{av}} = |E_0|^2/(2\eta)$ (W/m²)

LOSSY MEDIA

α — atten. const (Np/m)
β — phase const (rad/m)
$\gamma = \alpha + j\beta$ — prop. const
η_c — complex impedance (Ω)
$\delta_s = 1/\alpha$ — skin depth (m)
$\varepsilon' = \varepsilon_r\varepsilon_0$; $\varepsilon'' = \sigma/\omega$

CIRCULAR WAVES

RHC: $\tilde{\mathbf{E}} = a(\hat{x} - j\hat{y})e^{-jkz}$
LHC: $\tilde{\mathbf{E}} = a(\hat{x} + j\hat{y})e^{-jkz}$
Linear: $a_R = a_L = a_x/2$
$-j = e^{-j\pi/2}$: cos $\to$ sin
$+j = e^{+j\pi/2}$: cos $\to$ − sin

POLARIZATION

Linear: $\delta = 0$ or π
LHC: $a_x = a_y$, $\delta = +\pi/2$
RHC: $a_x = a_y$, $\delta = -\pi/2$
$R = 1$: circular; $R \to \infty$: linear
$\chi > 0$: RH; $\chi < 0$: LH